

Logical Agent Case Studies – Project Report

Abstract

This project analyzes four logical-agent case studies using propositional logic, FOL, forward chaining, backward chaining, unification, CNF conversion, and resolution.

Objectives

Represent knowledge formally; apply forward/backward chaining; compute MGUs; convert rules to CNF; prove goals using resolution; evaluate logical sufficiency and safety.

Methodology

Each case is translated into logical expressions. Rules are converted to inference-ready forms, facts are added, and the requested inference method is applied. Resolution uses proof by contradiction by adding the negated goal.

Results

Case 1 derives Possible Flu and Possible Pneumonia. Case 2 proves Proceed(CarA). Case 3 proves ApplyDripMethod; removing SoilDry breaks the proof. Case 4 derives Gold[2,2] but cannot prove the proposed path safe.

Conclusion

The cases demonstrate that logical agents can derive useful conclusions when rules and facts are explicit. They also show that absence of a fact is not automatically its negation and that simplified knowledge bases may be insufficient for safety or multi-agent coordination.

Future Enhancements

Add automated FOL parsing, graphical proof trees, richer Wumpus constraints, emergency priority scheduling, route-conflict detection, and interactive interfaces.